

Power BI Finance Dashboard

Indian Banking & Financial Services Project

This guide walks you through all steps to build a 6-page professional Finance Dashboard in Power BI using the four datasets. Follow the phases in sequence — each build on the previous one.

Overview – Dashboard Pages & Build Phases

Page	Report Name	Key Visuals
1	Executive Summary	KPI Cards, Monthly Trend Line, State Map, Segment Donut
2	Transaction Analysis	Product Column Chart, Waterfall, Channel Matrix, Drill-through
3	Budget vs Actual	KPI Bar Chart, Variance Table, Conditional Formatting
4	Customer Insights	Age Histogram, KYC Donut, Credit Score Bands, State Chart
5	Time Intelligence	YoY Line, MoM Bar, YTD vs PYTD, Running Total
6	Scenario Analysis	What-If Slicer, Forecast Line, Budget Planning, RLS Demo

Build Phases

#	Phase	Covers
A	Connect & Load Data	Import 4 CSV files, set data types
B	Clean in Power Query	All data quality issues in each table
C	Build Data Model	Relationships, Star Schema, Calendar table
D	Write DAX Measures	Basic, Budget, Time Intelligence, Advanced
E	Build Report Pages	All 6 pages with visuals, slicers, tooltips
F	Advanced Features	What-If, RLS, Forecast, Publish

Phase A – Connect & Load Data

Import all four CSV files into Power BI Desktop before any transformation begins.

STEP 1 Open Power BI Desktop & Save File

1. Open Microsoft Power BI Desktop on your machine.
2. Click New if Power BI opens with the start screen or go to File > New.
3. Immediately save: File > Save As > name it Finance_Dashboard_IB_and_FS.pbix

STEP 2 Import dim_customers.csv

4. Click Home tab on the ribbon.
5. Click Get Data > Text/CSV.
6. Browse to the file location, select dim_customers.csv, click Open.
7. Power BI shows a preview. Verify Customer_ID appears in the first column.

Do NOT click Load. Click Transform Data to open Power Query Editor.

IMPORTANT: Always click Transform Data (not Load) for all 4 files. This lets you clean data before it enters the model.

STEP 3 Import the Remaining Three Files

Inside Power Query Editor, go to Home > New Source > Text/CSV for each file:

- dim_customers.csv
- fact_transactions.csv
- fact_budget_actual.csv
- dim_products_loans.csv

After importing all four, you will see four queries in the Queries pane on the left.

STEP 4 Rename Queries for Clarity

In the Queries pane, right-click each query and rename as follows:

Original Name	Rename To
dim_customers	DimCustomers
fact_transactions	FactTransactions
fact_budget_actual	FactBudgetActual
dim_products_loans	DimProducts

TIP: Naming tables with Dim/Fact prefix is an industry standard. It makes the model view and DAX formulas much easier to read.

Phase B – Data Cleaning in Power Query Editor

Every step below maps to a specific Power Query technique. Work through each table in order.

B1 – Cleaning DimCustomers (500 rows)

STEP 5 Trim & Clean Customer_Name

8. Select the Customer_Name column (click the header).
9. Right-click the column header > Transform > Trim (removes leading/trailing spaces).
10. Right-click again > Transform > Capitalize Each Word (ensures proper case).

STEP 6 Standardize Gender Column

11. Select the Gender column.
12. Home > Replace Values: Enter M in Value to Find, Male in Replace With. Click OK.
13. Repeat for: F to Female, male to Male, FEMALE to Female, MALE to Male, female to Female.
14. Right-click Gender > Transform > Trim.

STEP 7 Fix Annual_Income_INR – Remove Rs. Prefix & Set Type

15. Select Annual_Income_INR column.
16. Transform > Replace Values: Rs. (with dot) to empty string. Click OK.
17. Right-click column header > Change Type > Decimal Number.

STEP 8 Parse Customer_Since with Mixed Date Formats

18. Select Customer_Since column.
19. Right-click > Change Type > Using Locale...
20. In the dialog: Data Type = Date, Locale = English (India). Click OK.

STEP 9 Standardize KYC_Status

21. Select KYC_Status. Transform > Trim.
22. Replace Values: verified to Verified, VERIFIED to Verified, Pending (with trailing space) to Pending.
23. Right-click > Transform > Capitalize Each Word.
24. Verify distinct values: right-click column > Group By to count unique values — should be exactly 4.

STEP 10 Validate Mobile Number Length

25. Add Column > Custom Column. Name it Mobile_Length.

```
= Text.Length(Text.From([Mobile]))
```

26. Filter Mobile_Length: Home > Filter Rows > does not equal 10.
27. Observe anomaly rows (leading zeros, extra digits) and correct the source data to 10 digits
28. Delete the Mobile_Length helper column after review.

STEP 11 Handle Blank Age & Credit_Score

29. Select Age column > Replace Values where null with 0
30. Repeat same process for Credit_Score (replace null with 0).
31. Change type of both new columns to Whole Number.

B2 – Cleaning FactTransactions (~5,100 rows)**STEP 12 Remove Duplicate Transaction_IDs**

32. Select the Transaction_ID column.
33. Home > Remove Rows > Remove Duplicates.

STEP 13 Parse Transaction_Date (Multiple Formats)

34. Select Transaction_Date column.
35. Right-click > Change Type > Using Locale > Date, English (India).
36. Verify by sorting: check that dates range correctly from 2010 to 2025.

STEP 14 Standardize Status Column

37. Select Status column. Transform > Trim.
38. Transform > Capitalize Each Word.
39. Replace Values: Success to Completed, under review to Under Review (fix case/spaces).
40. Filter to check remaining distinct values: Completed, Pending, Failed, Reversed, Under Review.

STEP 15 Fix Amount_INR, GST_INR, Net_Amount_INR, Processing_Fee_INR

41. For each money column, repeat these three actions:
 - ✓ Replace Values: Rs. to empty string (removes the prefix)
 - ✓ Transform > Trim (removes any remaining spaces)
 - ✓ Change Type > Decimal Number
42. Replace Errors > 0 for any rows that still fail conversion.

STEP 16 Add Conditional Column – High_Value_Flag

43. Add Column > Conditional Column. Name it High_Value_Flag.
44. Rule: If Net_Amount_INR is greater than 1000000 then High Value, else Normal.
45. Click OK. Change type to Text if needed.

B3 – Cleaning FactBudgetActual (1,200 rows)

STEP 17 Standardize Month_Name Column – Key Cleaning Demo

This column has four different formats: January, Jan, 1, and JANUARY. Standardize to full English name.

46. Add Column > Custom Column. Name it Month_Name_Clean.

```
= if Text.Lower([Month_Name]) = "january" or [Month_Name] = "jan" or [Month_Name] = "1"
then "January"
else if Text.Lower([Month_Name]) = "february" or [Month_Name] = "feb" or [Month_Name] =
"2" then "February"
else if Text.Lower([Month_Name]) = "march" or [Month_Name] = "mar" or [Month_Name] =
"3" then "March"
else if Text.Lower([Month_Name]) = "april" or [Month_Name] = "apr" or [Month_Name] =
"4" then "April"
else if Text.Lower([Month_Name]) = "may" or [Month_Name] = "5"
then "May"
else if Text.Lower([Month_Name]) = "june" or [Month_Name] = "jun" or [Month_Name] =
"6" then "June"
else if Text.Lower([Month_Name]) = "july" or [Month_Name] = "jul" or [Month_Name] =
"7" then "July"
else if Text.Lower([Month_Name]) = "august" or [Month_Name] = "aug" or [Month_Name] =
"8" then "August"
else if Text.Lower([Month_Name]) = "september" or [Month_Name] = "sep" or [Month_Name] =
"9" then "September"
else if Text.Lower([Month_Name]) = "october" or [Month_Name] = "oct" or [Month_Name] =
"10" then "October"
else if Text.Lower([Month_Name]) = "november" or [Month_Name] = "nov" or [Month_Name] =
"11" then "November"
else if Text.Lower([Month_Name]) = "december" or [Month_Name] = "dec" or [Month_Name] =
"12" then "December"
else [Month_Name]
```

47. Delete original Month_Name column. Rename Month_Name_Clean to Month_Name.

STEP 18 Fix Variance_Pct – Calculate the correct Variance Pct

48. Add Column > Custom Column > Rename Column name to Variance_Amount > Formula=

[Budget_INR_Lakhs]-[Actual_INR_Lakhs]
49. Add Column > Custom Column > Rename Column name to Variance_Amount_Pct >

[Variance_Amount_Pct]/[Budget_INR_Lakhs]*100

STEP 19 Standardize Target_Achieved Column

50. Replace Values: YES to Yes, yes to Yes, Y to Yes.
51. Replace Values: NO to No, N to No. Then Trim to remove trailing spaces.
52. Add Conditional Column Achievement_Flag: If Target_Achieved equals Yes then 1, else 0.
53. Change Achievement_Flag type to Whole Number (useful for DAX SUM later).

STEP 20 Handle Blank Year & Month

54. Sort table by Record_ID ascending (click the column header, select Sort Ascending) and check there are no blanks.

B4 – Cleaning DimProducts (100 rows)**STEP 21 Standardize Status, Risk_Category, Prepayment_Allowed, GST_Applicable**

Apply Replace Values and Capitalize Each Word to each column:

- ✓ Status: active to Active, ACTIVE to Active, discontinued to Discontinued.
- ✓ Risk_Category: LOW to Low, low to Low, HIGH to High, high to High.
- ✓ Prepayment_Allowed: YES to Yes, NO to No, yes to Yes, no to No.
- ✓ GST_Applicable: Applicable to Yes, N/A to No, yes to Yes.

STEP 22 Handle Blank Interest_Rate_Pct

55. Select Interest_Rate_Pct. Right-click > Replace Errors > 0 if any.
56. Home > Replace Values: null to 0.
57. Change Type > Decimal Number.

STEP 23 Add Loan_Band Conditional Column

58. Add Column > Conditional Column. Name it Loan_Band.
59. Create the following rules in order:
 - If Max_Loan_Amount_INR is less than 500000 then Micro
 - Else if Max_Loan_Amount_INR is less than 5000000 then Small
 - Else if Max_Loan_Amount_INR is less than 50000000 then Medium
 - Else Large

STEP 24 Close & Apply All Queries

60. Review all four queries in the Queries pane. Ensure no red error indicators are visible.
61. Go to Home > Close & Apply.
62. Power BI loads all cleaned data into the model. Watch the status bar at the bottom.
63. After loading, go to Data view. Spot-check values in each table.

NOTE: CHECKPOINT: After Close & Apply you should see approximately: DimCustomers ~500 rows, FactTransactions ~4,900 rows (duplicates removed), FactBudgetActual 1,200 rows, DimProducts 100 rows.

Phase C – Building the Data Model

STEP 25 Switch to Model View

64. Click the Model view icon on the left sidebar (looks like three connected boxes).
65. You will see all four tables displayed as boxes.
66. Arrange: place DimCustomers and DimProducts on the outer ring, the two Fact tables in the centre.

STEP 26 Create Relationship: FactTransactions to DimCustomers

67. Drag the Customer_ID field from FactTransactions and drop it onto Customer_ID in DimCustomers.
68. In the Create Relationship dialog, verify: Cardinality = Many to One, Cross Filter = Single.
69. Click OK.

STEP 27 Create Relationship: FactTransactions to DimProducts

70. Drag Product column from FactTransactions onto Product_Name column in DimProducts.
71. Cardinality: Many to One. Cross Filter Direction: Single. Click OK.

NOTE: The Product column in FactTransactions stores the product name as text, not a Product_ID. This is a text-based relationship — a very common real-world scenario to teach.

STEP 28 Create DAX Calendar Table

72. Click Modeling tab > New Table.
73. Enter the following DAX expression and press Enter:

```
Calendar = CALENDAR(DATE(2010, 1, 1), DATE(2025, 12, 31))
```

74. A new Calendar table appears in the Data pane with a single [Date] column.

STEP 29 Add Calculated Columns to the Calendar Table

Click on the Calendar table in Data view. For each column below, click Modelling > New Column:

Year	= YEAR([Date])
Month Number	= MONTH([Date])
Month Name	= FORMAT([Date], "MMMM")
Month Short	= FORMAT([Date], "MMM")
Quarter	= "Q" & ROUNDUP(MONTH([Date])/3, 0)
Day Name	= FORMAT([Date], "DDDD")
Week Number	= WEEKNUM([Date])
Is Weekend	= IF(WEEKDAY([Date],2)>5, TRUE, FALSE)
Financial Year	= IF(MONTH([Date])>=4, "FY" & YEAR([Date]) & "-" & RIGHT(YEAR([Date])+1,2), "FY" & YEAR([Date])-1 & "-" & RIGHT(YEAR([Date]),2))

STEP 30 Mark Calendar as Date Table & Link to Facts

75. Right-click Calendar table in the Data pane > Mark as Date Table.
76. Select [Date] column as the date column. Click OK.

77. In Model view, drag Date from Calendar onto Transaction_Date in FactTransactions.
78. Drag Date from Calendar onto Last_Updated in FactBudgetActual.
79. Verify both relationships show as active.

IMPORTANT: The Calendar table MUST be marked as Date Table before Time Intelligence functions like TOTALYTD and SAMEPERIODLASTYEAR will work correctly.

Phase D – Writing DAX Measures

Create a dedicated container table to hold all measures. Click Modelling > New Table and enter:

```
MeasuresTable = ROW("Info", "Finance Dashboard Measures")
```

TIP: Right-click MeasuresTable in the Data pane > Hide in Report View. It is a container only — all measures created here will still appear in visuals.

D1 – Basic Transaction Measures (Module 4)

Measure Name	DAX Formula
Total Disbursement	SUM(FactTransactions[Amount_INR])
Total Transactions	COUNTROWS(FactTransactions)
Avg Transaction Value	AVERAGE(FactTransactions[Amount_INR])
Total Customers	DISTINCTCOUNT(FactTransactions[Customer_ID])
Total GST Collected	SUM(FactTransactions[GST_INR])
Total Processing Fee	SUM(FactTransactions[Processing_Fee_INR])
Net Revenue	SUM(FactTransactions[Net_Amount_INR])
High Value Count	CALCULATE(COUNTROWS(FactTransactions), FactTransactions[Amount_INR]>1000000)

D2 – Budget vs Actual Measures

Measure Name	DAX Formula
Total Budget	SUM(FactBudgetActual[Budget_INR_Lakhs])
Total Actual	SUM(FactBudgetActual[Actual_INR_Lakhs])
Variance	[Total Actual] - [Total Budget]
Variance %	DIVIDE([Variance], [Total Budget], 0)
Budget Achievement %	DIVIDE([Total Actual], [Total Budget], 0)
Targets Met	CALCULATE(COUNTROWS(FactBudgetActual), FactBudgetActual[Target_Achieved]="Yes")
Variance Flag	IF([Variance]>=0, "Over Target", "Under Target")

D3 – Time Intelligence Measures (Module 7 – Key Focus)

Measure Name	DAX Formula
YTD Disbursement	TOTALYTD(SUM(FactTransactions[Amount_INR]), Calendar[Date])
PYTD Disbursement	CALCULATE([Total Disbursement], SAMEPERIODLASTYEAR(Calendar[Date]))
YoY Growth	DIVIDE([Total Disbursement]-[PYTD Disbursement], [PYTD Disbursement], 0)
Last Month Amount	CALCULATE([Total Disbursement], DATEADD(Calendar[Date],-1,MONTH))
MoM Growth	DIVIDE([Total Disbursement]-[Last Month Amount],[Last Month Amount],0)
Running Total	CALCULATE([Total Disbursement], DATESYTD(Calendar[Date]))
QTD Disbursement	TOTALQTD(SUM(FactTransactions[Amount_INR]), Calendar[Date])
Last Year YTD	CALCULATE([YTD Disbursement], SAMEPERIODLASTYEAR(Calendar[Date]))

IMPORTANT: Before testing Time Intelligence measures, confirm: (1) Calendar table is marked as Date Table, (2) Calendar[Date] is linked to Transaction_Date in FactTransactions. Without these, TOTALYTD and SAMEPERIODLASTYEAR will not work.

D4 – Advanced DAX (CALCULATE, FILTER, ALL)

Measure Name	DAX Formula
Home Loan Revenue	CALCULATE([Total Disbursement], FactTransactions[Product]="Home Loan")
South Zone Revenue	CALCULATE([Total Disbursement], FactTransactions[Branch_Name]="South Zone")
% of Total Revenue	DIVIDE([Total Disbursement], CALCULATE([Total Disbursement],ALL(FactTransactions)),0)
Completed Txns Only	CALCULATE([Total Disbursement], FILTER(FactTransactions, FactTransactions[Status]="Completed"))
Safe Division Demo	DIVIDE([Total Disbursement],[Total Transactions],0)

Phase E – Building All 6 Report Pages

Set the canvas theme before creating any pages. Go to View tab > Themes > Executive for a professional finance look.

Page 1 – Executive Summary Dashboard

STEP 31 Create KPI Cards – Top Row

80. Visualizations pane > click the Card visual icon.
81. Drag [Total Disbursement] to the Fields well. In Format pane, set Display Units to Crores.
82. Copy the card (Ctrl+C, Ctrl+V) and replace the measure with:
 - Card 2: Total Transactions
 - Card 3: Total Customers
 - Card 4: Budget Achievement %
 - Card 5: Net Revenue
83. Arrange all 5 cards in a horizontal row across the top of the canvas.

STEP 32 Create Monthly Trend Line Chart

- 84. Visualizations pane > Line Chart.
- 85. X-axis: Calendar[Month Name]. Sort by Month Number (click ... on visual > Sort Axis).
- 86. Y-axis: [Total Disbursement].
- 87. Legend: FactTransactions[Branch_Name].
- 88. Format > Title: Monthly Disbursement by Branch Zone.
- 89. Enable Analytics pane > Average Line. Turn on Data Labels.

STEP 33 Create State-wise Filled Map

- 90. Visualizations > Filled Map.
- 91. Location: DimCustomers[State]. Set Data Category = State or Province.
- 92. Values: [Total Customers]. Color Saturation: [Total Disbursement].
- 93. Format > Map styles > Color theme: Sequential (white to dark blue).

TIP: If states do not show correctly, add a Country column to DimCustomers with value India and set its Data Category to Country.

STEP 34 Add Segment Donut Chart

- 94. Visualizations > Donut Chart.
- 95. Legend: DimCustomers[Segment]. Values: [Total Customers].
- 96. Format > Detail Labels: On, show percentage and value.

STEP 35 Add Slicers to Page 1

- 97. Visualizations > Slicer. Field: Calendar[Year]. Style: Dropdown.
- 98. Add second Slicer: Calendar[Quarter].
- 99. Add third Slicer: DimCustomers[State].
- 100. Add fourth Slicer: DimCustomers[Segment].
- 101. Arrange slicers in a left panel or horizontal strip at the top.

Page 2 – Transaction Analysis**STEP 36 Product-wise Clustered Column Chart**

- 102. Right-click page tab at bottom > Add Page. Name it Transaction Analysis.
- 103. Visualizations > Clustered Column Chart.
- 104. X-axis: FactTransactions[Product]. Y-axis: [Total Disbursement].
- 105. Legend: FactTransactions[Transaction_Type].
- 106. Sort descending by Total Disbursement: click ... on visual > Sort Axis > Total Disbursement Descending.

STEP 37 Create Waterfall Chart – Monthly Net Revenue

- 107. Visualizations > Waterfall Chart.
- 108. Category: Calendar[Month Name]. Y-axis: [Net Revenue].
- 109. Breakdown: FactTransactions[Branch_Name].

110. Format > Sentiment Colors: Increase = Green, Decrease = Red, Total = Dark Blue.

STEP 38 Channel-wise Matrix with Conditional Formatting

111. Visualizations > Matrix.
 112. Rows: FactTransactions[Channel]. Columns: Calendar[Year]. Values: [Total Disbursement].
 113. Enable Conditional Formatting: click Values well > Format > Conditional Formatting > Background Color > Gradient.
 114. Set: low = white, high = dark blue. This creates a heat map effect on the matrix.

STEP 39 Set Up Drill-through to Transaction Detail Page

115. Add a new page named Transaction Detail — this is the drill-through target page.
 116. Add a Table visual: columns = Transaction_ID, Transaction_Date, Product, Amount_INR, Status, RM_Name.
 117. In Visualizations > Format > Drill-through: drag FactTransactions[Product] to the Drill-through field.
 118. Go back to Transaction Analysis page. Right-click any bar > Drill through > Transaction Detail.

Page 3 – Budget vs Actual

STEP 40 KPI Comparison Horizontal Bar Chart

119. Add new page named Budget vs Actual.
 120. Visualizations > Clustered Bar Chart.
 121. Y-axis: FactBudgetActual[KPI_Name]. X-axis: add both [Total Budget] and [Total Actual].
 122. Format > Data Labels: On. Show Legend at bottom.

STEP 41 Variance Table with Conditional Formatting

123. Visualizations > Table.
 124. Columns: Branch_Zone, KPI_Name, [Total Budget], [Total Actual], [Variance], [Variance %].
 125. Format [Variance %] as Percentage with 1 decimal place.
 126. Apply Conditional Formatting on Variance: click Values well ... > Conditional Formatting > Font Color.
 127. Rules: If value is less than 0 then Red. If value is greater than or equal to 0 then Green.
 128. Apply the same rules on the Variance % column.

STEP 42 Add Year, KPI and Zone Slicers

129. Add Slicer: Calendar[Year] (Dropdown style).
 130. Add Slicer: FactBudgetActual[KPI_Name] (List style so all KPIs are visible simultaneously).
 131. Add Slicer: FactBudgetActual[Branch_Zone].
 132. Demo: Selecting a single KPI (e.g., Total Disbursement) updates both chart and table instantly.

Page 4 – Customer Insights

STEP 43 Age Group Histogram Using Binning

- 133. In Data view, right-click DimCustomers[Age] > New Group.
- 134. Group type: Bin. Bin size: 10. Minimum: 20, Maximum: 70.
- 135. Rename the group column to Age_Group.
- 136. Add a Clustered Column Chart: X-axis = Age_Group, Y-axis = [Total Customers].

STEP 44 KYC Status Donut Chart

- 137. Visualizations > Donut Chart.
- 138. Legend: DimCustomers[KYC_Status]. Values: Count of Customer_ID.
- 139. Format: Show percentage labels. Custom colors: Verified=Green, Pending=Orange, Expired=Red.

STEP 45 Credit Score Band Bar Chart

- 140. In Data view, add Calculated Column to DimCustomers (Modeling > New Column):

```
Credit_Score_Band =
IF(DimCustomers[Credit_Score]<500, "Poor (<500)",
IF(DimCustomers[Credit_Score]<650, "Fair (500-649)",
IF(DimCustomers[Credit_Score]<750, "Good (650-749)",
IF(DimCustomers[Credit_Score]<850, "Very Good (750-849)",
"Excellent (850+)"))))
```

- 141. Add Column Chart: X = Credit_Score_Band, Y = Count of Customer_ID.
- 142. Sort bands in logical order using Modeling > Sort by Column feature.

Page 5 – Time Intelligence**STEP 46 YoY Growth Line Chart**

- 143. Add new page named Time Intelligence.
- 144. Visualizations > Line Chart.
- 145. X-axis: Calendar[Year]. Y-axis: add [Total Disbursement] and [PYTD Disbursement] as two lines.
- 146. Format > Legend: Show, position Bottom. Turn on Data Labels.
- 147. Add a KPI Card below showing [YoY Growth] formatted as a percentage.

STEP 47 Month-on-Month Bar Chart

- 148. Visualizations > Clustered Bar Chart.
- 149. Y-axis: Calendar[Month Name] (sort by Month Number). X-axis: [Total Disbursement].
- 150. Add [Last Month Amount] as a second series for side-by-side comparison.

STEP 48 YTD vs PYTD KPI Visual

- 151. Visualizations > KPI Visual.
- 152. Value: [Total Disbursement]. Trend axis: Calendar[Month Name]. Target: [PYTD Disbursement].
- 153. The KPI visual shows a green arrow when current exceeds prior year, red arrow when below.

STEP 49 Running Total Area Chart

- 154. Visualizations > Area Chart.

- 155. X-axis: Calendar[Date] set to Month level using the drill-down hierarchy.
- 156. Y-axis: [Running Total].
- 157. Format > Fill: Light blue gradient. Title: Cumulative Disbursement – Year to Date.

Page 6 – Scenario Analysis & Advanced Features

STEP 50 Create What-If Parameter

- 158. Modeling tab > New Parameter (What If Parameter in older versions).
- 159. Minimum: 0, Maximum: 0.30, Increment: 0.01, Default: 0.10. Click OK.
- 160. Power BI creates a slicer and a parameter measure automatically.

STEP 51 Add Analytics Pane Forecast

- 161. Click on the Monthly Trend Line Chart from Page 1 (or create a copy on this page).
- 162. In the Analytics pane (magnifying glass icon), click Forecast.
- 163. Forecast Length: 12 periods. Confidence Interval: 95%.
- 164. Show students the shaded forecast region and how it extends beyond the data.

STEP 52 Implement Row-Level Security (RLS)

- 165. Go to Modeling tab > Manage Roles.
- 166. Click Create. Name the role South_Zone_RM.
- 167. Select FactTransactions table. In the Table Filter DAX Expression, enter:

```
[Branch_Name] = "South Zone"
```

- 168. Click Save. Then click Modeling > View as Roles.
- 169. Select South_Zone_RM. Observe: only South Zone transactions appear in all visuals.
- 170. Click Stop Viewing to return to full data view.

NOTE: In real deployment, RLS roles are mapped to Azure Active Directory users or groups when the report is published to Power BI Service.

Phase F – Polish, Interactivity & Publishing

STEP 53 Enable Drill-Down on Time Visuals

- 171. Click on any line chart that uses Calendar hierarchy.
- 172. In the visual header, click the double-arrow drill-down icon to enable drill mode.
- 173. Click a data point to drill from Year to Quarter to Month to Day.
- 174. Demonstrate: the hierarchy on Calendar enables this automatically.

STEP 54 Configure Cross-Page Slicer Sync

- 175. View tab > Sync Slicers panel.
- 176. Select the Year slicer on Page 1.
- 177. In the Sync Slicers panel, check all 6 pages so the Year slicer applies everywhere.
- 178. Demo: selecting a Year on Page 1 automatically filters all other pages simultaneously.

STEP 55 Add Bookmarks & Navigation Buttons

- 179. View > Bookmarks pane. On each page, click Add to save the default state.
- 180. Insert > Buttons > Blank. In Format > Button Text: Executive Summary.
- 181. In Format > Action: Type = Bookmark. Bookmark = Page 1 bookmark.
- 182. Copy button and set it for each of the 6 pages. Align all buttons in a navigation strip.

STEP 56 Apply Conditional Formatting to KPI Cards

- 183. Click a KPI Card visual (e.g., YoY Growth).
- 184. Format pane > Callout value > Font Color > Conditional Formatting.
- 185. Rule: If YoY Growth is less than 0 then Red. If greater than or equal to 0 then Green.
- 186. Apply the same logic to Budget Achievement % card.

STEP 57 Publish to Power BI Service

- 187. File > Publish > Publish to Power BI.
- 188. Sign in with a Microsoft account (trainer account for demo).
- 189. Select workspace: My Workspace or a designated training workspace.
- 190. After publishing, click the link in the dialog to open the report in the browser.
- 191. In Power BI Service: click the report > Pin a visual to a Dashboard.
- 192. Create a new Dashboard named Finance KPI Dashboard.
- 193. Pin 4 to 5 key visuals from different report pages to the dashboard.

NOTE: Dashboard vs Report: A Report is multi-page and interactive, created in Power BI Desktop. A Dashboard is a single-page canvas in Power BI Service that pins tiles from reports. Both serve different audiences.

STEP 58 Set Up Data Refresh (Concept)

- 194. In Power BI Service, go to Dataset settings (click ... next to the dataset name).
- 195. Explain: since data is from CSV files, refresh requires re-uploading or using a Gateway.
- 196. Discuss real-world scenario: banks connect Power BI to SQL Server or Azure SQL, enabling automatic scheduled refresh (hourly or daily).

Student Project Submission Checklist

Each student must complete and be able to demonstrate all of the following:

#	Requirement
1	All 4 CSV files imported and cleaned in Power Query Editor
2	Star Schema created with all 5 relationships (FactTransactions-DimCustomers, FactTransactions-DimProducts, Calendar-FactTransactions, Calendar-FactBudgetActual)
3	Calendar table created with Year, Month, Quarter, Financial Year columns and marked as Date Table
4	Minimum 20 DAX measures created across all 4 categories: Basic, Budget, Time Intelligence, Advanced
5	All 6 report pages created with correct visuals and working interactions
6	Slicers working on Year, Quarter, State, Segment synced across pages
7	Drill-through from Transaction Analysis to Transaction Detail page working
8	Conditional formatting applied on Variance table (red for negative, green for positive)
9	Waterfall chart on Page 2 and KPI visual on Page 5 present and correct
10	What-If parameter created and Forecast Revenue measure working on Page 6
11	Row-Level Security role demonstrated (South_Zone_RM filters to South Zone only)
12	Report published to Power BI Service and at least 4 visuals pinned to a Dashboard